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1. Peano axioms

We use $\mathbb{N} = \{1, 2, 3, \dots\}$ and write $\hat{n}$ for the successor of n .

Axiom 1.1 (Peano axioms).

1. $1 \in \mathbb{N}$.
2. For every $n \in \mathbb{N}$, $\hat{n} \in \mathbb{N}$.
3. For every $n \in \mathbb{N}$, $\hat{n} \neq 1$.
4. $\hat{m} = \hat{n} \Leftrightarrow m = n$.
5. **Principle of induction:** if $P(1)$ is true and $P(n) \Rightarrow P(\hat{n})$ for every $n \in \mathbb{N}$, then $P(n)$ is true for every $n \in \mathbb{N}$.

1.1. Recursive Definition $+$ and $\cdot$

Definition 1.2. For $n, m \in \mathbb{N}$,

$$+ : \begin{cases} n + 1 = \hat{n} \\ n + \hat{m} = \widehat{n + m} \end{cases}$$
$$\cdot : \begin{cases} n \cdot 1 = n \\ n \cdot \hat{m} = n \cdot m + n \end{cases}$$

Note. There are definitions of $+$ and $\cdot$ not propositions

2. Addition: associativity and commutativity

2.1. Associative Law

Proposition 2.1. For all $a, b, c \in \mathbb{N}$,

$$(a + b) + c = a + (b + c).$$

Proof. Let $P(c) := (a + b) + c = a + (b + c)$ is true

Base case ($c = 1$):

$$(a + b) + 1 = \widehat{a + b} = a + \hat{b} = a + (b + 1).$$

Inductive step: suppose $P(c)$ holds. Then

$$\begin{aligned} (a + b) + \hat{c} &= (\widehat{(a + b)} + c) \text{ (recursive rule)} \\ &= (a + \widehat{(b + c)}) \text{ (inductive hypothesis)} \\ &= a + \widehat{b + c} \text{ (recursive rule)} \\ &= a + (b + \hat{c}) \text{ (recursive rule).} \end{aligned}$$

Thus $P(\hat{c})$ holds. By induction, addition is associative. ■

2.2. Commutative Law

Lemma 2.2. $a + 1 = 1 + a$ for all $a \in \mathbb{N}$

Proof. The base case is $1 + 1 = 1 + 1$. If $1 + a = a + 1$, then

$$1 + \hat{a} = 1 + (a + 1) = (1 + a) + 1 = (a + 1) + 1 = \hat{a} + 1. \quad \blacksquare$$

Remark. We used associative law and definitions here

Lemma 2.3. $\hat{a} + b = \widehat{a + b} = a + \hat{b}$

Proof. By associativity and the $+$ definitions,

$$\hat{a} + b = (a + 1) + b = a + (1 + b) = a + (b + 1) = a + \hat{b} = \widehat{a + b} \quad \blacksquare$$

Proposition 2.4. $a + b = b + a$ for all $a, b \in \mathbb{N}$

Proof. Fix a and induct on b . The base case $a + 1 = 1 + a$ is Lemma 2.2. If $a + b = b + a$, then

$$a + \hat{b} = \widehat{a + b} = \widehat{b + a} = \hat{b} + a.$$

By def of $+$, IH ■

Exercise 2.5. Prove that multiplication is associative and commutative using the Peano axioms and the recursive $\cdot$ definitions

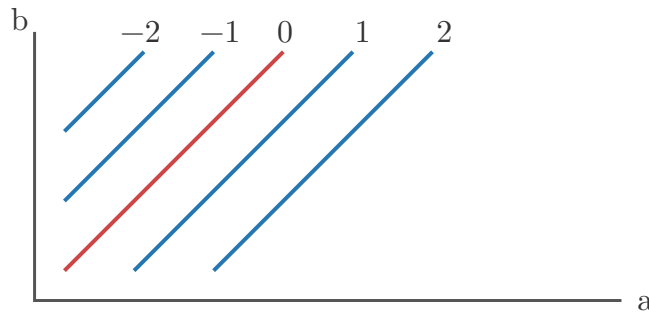
3. Construction of the integers

The idea of an equivalence relation is to identify different points. Here, we identify pairs along diagonals, and the difference between two coordinate would be what it identify to.

Definition 3.1. Define a relation on $\mathbb{N} \times \mathbb{N}$ by

$$(a, b) \sim (c, d) \quad \text{iff} \quad a + d = b + c.$$

Intuition: Think of (a, b) as the difference $a - b$. The relation says that $a - b = c - d$. We want two tuples with same difference to be equivalent.



Exercise 3.2. Prove that $\sim$ is an equivalence relation

Definition 3.3. Define $\mathbb{Z} := (\mathbb{N} \times \mathbb{N}) / \sim$

$$0 := [(1, 1)] = [(2, 2)], \quad n := [(n + 1, 1)], \quad -n := [(1, n + 1)]$$

3.1. Addition and multiplication on Integer

Definition 3.4. Define addition on equivalence classes by

$$[(a, b)] + [(c, d)] := [(a + c, b + d)].$$

Proposition 3.5. Addition on the integer equivalence classes is well-defined

Proof. Suppose $(a, b) \sim (a', b')$ and $(c, d) \sim (c', d')$ s.t. $a + b' = b + a'$ and $c + d' = d + c'$

$$\begin{aligned} (a + c) + (b' + d') &= (b + a') + (d + c') \\ &= (a' + c') + (b + d) \end{aligned}$$

Hence $(a + c, b + d) \sim (a' + c', b' + d')$ Therefore,

$$[(a, b)] + [(c, d)] = [(a + c, b + d)] = [(a' + c', b' + d')] = [(a', c')] + [(b', d')] \quad \blacksquare$$

Note. Change of representatives should not change the output

Definition 3.6. Define multiplication by

$$[(a, b)][(c, d)] := [(ac + bd, ad + bc)].$$

Note. Think as $(a - b)(c - d) = (ac + bd) - (bc + ad)$

Exercise 3.7. Check multiplication is well-defined

4. Construction of the rationals

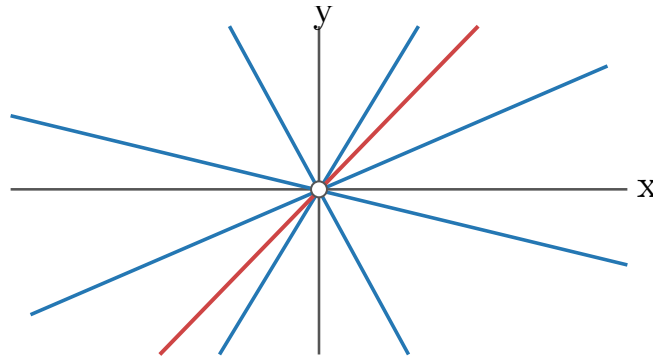
Definition 4.1. On $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, define

$$(a, b) \sim (c, d) \quad \text{iff} \quad ad = bc$$

Then

$$\mathbb{Q} := (\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})) / \sim$$

Exercise 4.2. Check that $\sim$ is an equivalence relation



Remark. Every points on the line are identify as the equivalent. The slope of the line is the rational number. This picture is similar to the real projective plane $\mathbb{RP}^2$ from the last week, but the domain for the real projective plane is $\mathbb{R}^2$ instead of integer grids

Definition 4.3. Define the operations by

$$[(a, b)] + [(c, d)] := [(ad + bc, bd)],$$

$$[(a, b)][(c, d)] := [(ac, bd)].$$

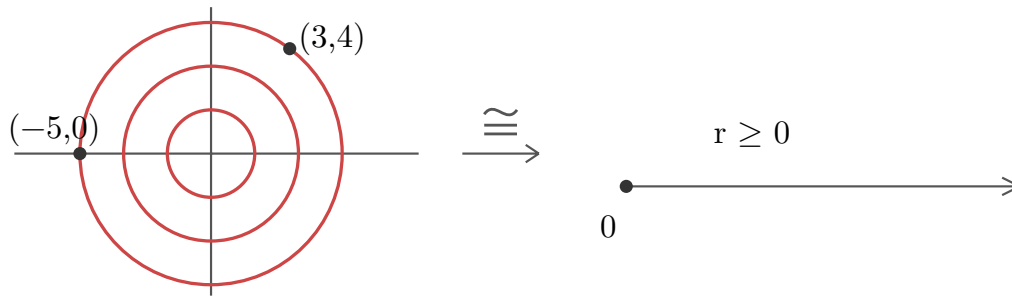
Exercise 4.4. Check that addition and multiplication are well-defined

4.1. Practice Problems from Week 2 Notes

(Week 2, 1st problem) For $p, q \in \mathbb{R}^2$, define $p \sim q$ iff $\|p\|_2 = \|q\|_2$. We can calculate them directly,

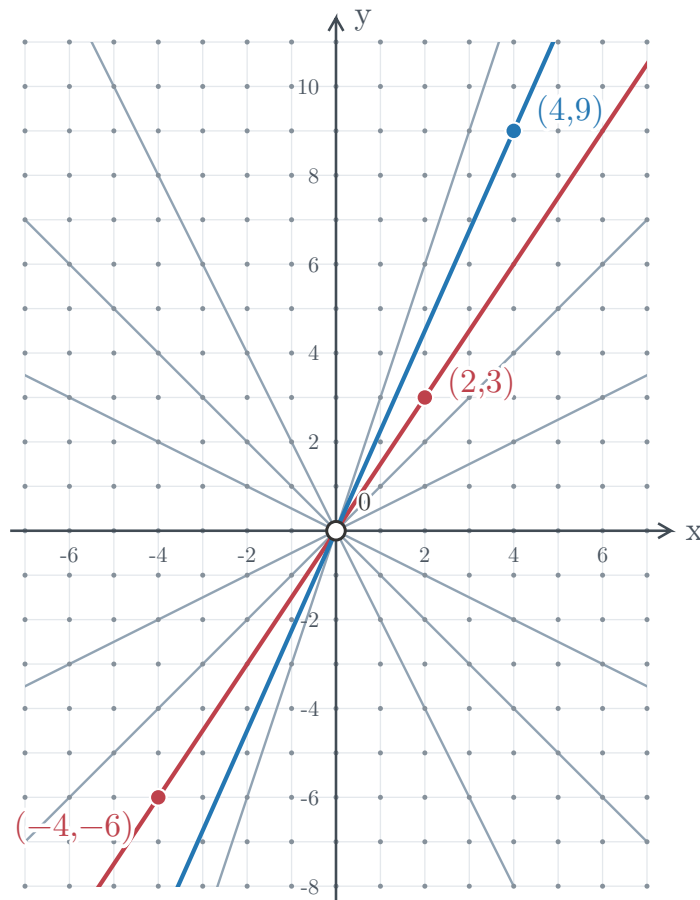
$$\|(3, 4)\|_2 = \sqrt{3^2 + 4^2} = 5 = \sqrt{(-5)^2 + 0^2} = \|(-5, 0)\|_2,$$

so $(3, 4) \sim (-5, 0)$.



Each class is a circle centered at the origin (or just the origin when the radius is zero). Choose $(\sqrt{x^2 + y^2}, 0)$, as its representative. Thus the quotient is the non-negative real number line.

(Week 2, 6th problem)



The open circle excludes the origin. The points $(2, 3)$ and $(-4, -6) = -2(2, 3)$ lie on the same line $y = \frac{3}{2}x$, so they represent the same projective direction. The point $(4, 9)$ lies on the distinct line $y = \frac{9}{4}x$.

Unlike the previous problem, we cannot find a more pleasant representatives than having the minimal fraction to represent the line. It will look like a bunch of points that never line up together.

5. Ordering

Definition 5.1. Define relations on $\mathbb{N}$ by

$$p < q \Leftrightarrow \exists m \in \mathbb{N} : p + m = q,$$

$$p \leq q \Leftrightarrow p = q \text{ or } \exists m \in \mathbb{N} : p + m = q.$$

5.1. Irreflexivity of $<$

Proposition 5.2. *For every $a \in \mathbb{N}$, $a < a$ is false*

Proof. We prove the equivalent statement by induction:

$$P(a) : \quad \forall d \in \mathbb{N}, \quad a + d \neq a.$$

Base case: $1 + d = d + 1 = \hat{d} \neq 1$, since 1 is not a successor.

Inductive step: suppose $P(a)$ holds and let $d \in \mathbb{N}$ be arbitrary. Then

$$\hat{a} + d = \widehat{a + d} \neq \hat{a},$$

since $a + d \neq a$ and Peano axiom. ■

5.2. Transitivity of $<$

Proposition 5.3. *If $a < b$ and $b < c$, then $a < c$*

Proof. Suppose $a < b$ and $b < c$. There exist $m_1, m_2 \in \mathbb{N}$ with

$$a + m_1 = b, \quad b + m_2 = c.$$

Therefore

$$(a + m_1) + m_2 = c \quad \Rightarrow \quad a + (m_1 + m_2) = c.$$

Since $m_1 + m_2 \in \mathbb{N}$, we have $a < c$. ■

5.3. Immediate successor

Proposition 5.4. *For each $n \in \mathbb{N}$, $n < \hat{n}$, and there is no $x \in \mathbb{N}$ such that*

$$n < x < \hat{n}.$$

Proof. The first statement follows from $n + 1 = \hat{n}$. For the second, suppose such an x exists. Then

$$n + d_1 = x, \quad x + d_2 = \hat{n} \quad (d_1, d_2 \in \mathbb{N}).$$

It follows that

$$n + (d_1 + d_2) = n + 1 \quad \Rightarrow \quad d_1 + d_2 = 1,$$

by cancellation. This is impossible: a sum of two positive natural numbers is a successor and therefore cannot equal 1. ■

Exercise 5.5. Prove the cancellation law for addition from the Peano axioms

5.4. Strict Totality of $<$

Proposition 5.6. *For distinct $a, b \in \mathbb{N}$, either $a < b$ or $b < a$. More precisely, exactly one of*

$$a < b, \quad a = b, \quad b < a$$

holds.

Proof. First, every natural number is either 1 or a successor: this follows by induction, since the base case is 1 and $\hat{n}$ is a successor for every $n \in \mathbb{N}$.

Fix $a \in \mathbb{N}$ and induct on b . Let $P(b)$ be the statement that $a < b$, $a = b$, or $b < a$.

Base case ($b = 1$): if $a = 1$, we have equality. Otherwise $a = \hat{c} = c + 1 = 1 + c$ for some $c \in \mathbb{N}$, so $1 < a$.

Inductive step: suppose $P(b)$ holds. We compare a with $\hat{b}$ in each case.

- If $a < b$, then $a < \hat{b}$ by transitivity, since $b < \hat{b}$.

- If $a = b$, then $a < \hat{b}$ by the immediate-successor property.
- If $b < a$, there is a $d \in \mathbb{N}$ such that $b + d = a$. If $d = 1$, then $a = b + 1 = \hat{b}$. Otherwise $d = \hat{e}$ for some $e \in \mathbb{N}$, and

$$a = b + \hat{e} = \widehat{b + e} = \hat{b} + e.$$

Hence $\hat{b} < a$.

Thus $P(\hat{b})$ holds ■

Remark. For a strict order, totality compares distinct elements. Together with irreflexivity and transitivity, this shows that $<$ is a strict total order on $\mathbb{N}$.

6. Total and Partial orderings

6.1. Total ordering

Definition 6.1. A relation $\leq$ is a **total order** if it satisfies all three properties:

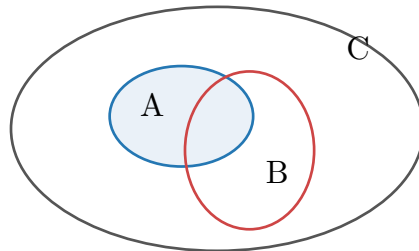
1. **Reflexivity:** $a \leq a$.
2. **Antisymmetry:** if $a \leq b$ and $b \leq a$, then $a = b$.
3. **Totality:** for every pair a, b , either $a \leq b$ or $b \leq a$.

Definition 6.2. A relation $\leq$ is a **partial order** if it satisfies all three properties:

1. **Reflexivity:** $a \leq a$.
2. **Antisymmetry:** if $a \leq b$ and $b \leq a$, then $a = b$.
3. **Transitivity:** if $a \leq b$ and $b \leq c$, then $a \leq c$.

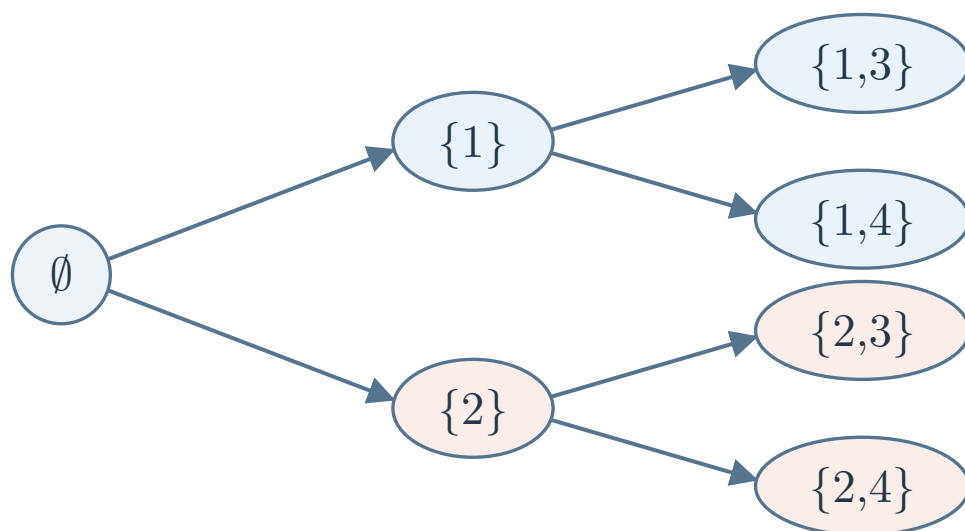
6.2. Example of a partial ordering: subsets

Example. Subset inclusion $\subseteq$ is a partial order on the power set of a set: it is reflexive, antisymmetric, and transitive. It need not be total, since two subsets may be incomparable.



For example, $A = \{1, 2\}$ and $B = \{2, 3\}$ are both subsets of $C = \{1, 2, 3\}$, but $A \not\subseteq B$ and $B \not\subseteq A$.

The following tree shows subset inclusion on the displayed family of sets. Each arrow points from a smaller set to a larger set.



For example, $\emptyset \subseteq \{1\} \subseteq \{1, 3\}$, so $\emptyset \subseteq \{1, 3\}$ by transitivity. However, $\{1\}$ and $\{2\}$ are incomparable, as are $\{1, 3\}$ and $\{1, 4\}$: neither set is a subset of the other.

Note. Subset inclusion is reflexive, antisymmetric, and transitive. On this family it is a partial order, but not a total order.